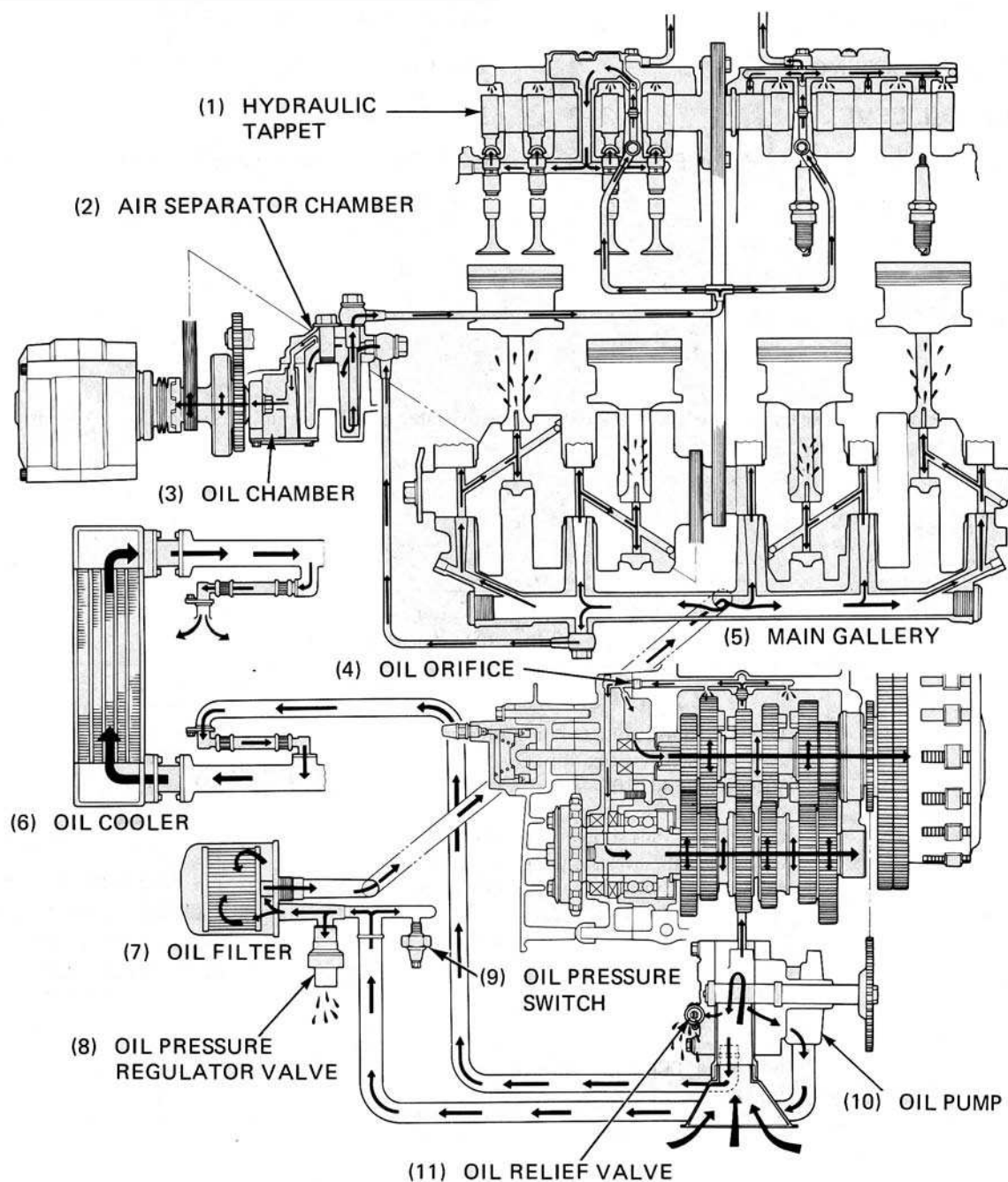




HONDA CBX750F

2. Lubrication





HONDA CBX750F

2. Lubrication

SERVICE INFORMATION	2-1	OIL STRAINER/RELIEF VALVE	2-5
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ENGINE OIL & FILTER CHANGE	2-3	CONTROL CABLE LUBRICATION	2-13
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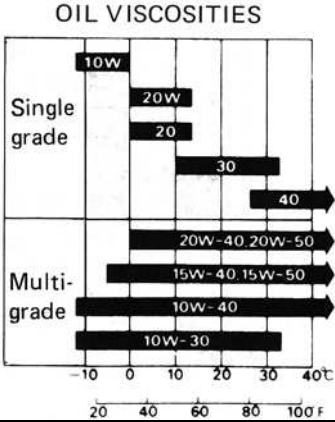
SERVICE INFORMATION

GENERAL

To service the oil pump, it is necessary to remove the exhaust system and clutch assembly. See section 8 for clutch removal and installation.

SPECIFICATIONS

Engine oil

Oil capacity	2.8 litre (3.0 U.S. qt, 2.5 Imp. qt) after oil filter and oil change 3.6 litre (3.8 U.S. qt, 3.2 Imp. qt) after disassembly	
Oil recommendation	Use Honda 4-Stroke Oil or equivalent. API Service Classification: SE or SF. Viscosity: SAE 10W-40 Other viscosities shown in the chart may be used when the average temperature in your riding area is within the indicated range.	
Oil pressure (at oil pressure switch)	630 kPa/6,000 min' 80°C (176° F) (6.3 kg/cm ² /6,000 rpm, 90 PSI/7,000 rpm)	

Oil pump service data

	STANDARD	SERVICE LIMIT
Rotor tip clearance	0.10 mm (0.004 in)	0.15 mm (0.006 in)
Pump body clearance	0.15-0.22 mm (0.006-0.009 in)	0.35 mm (0.014 in)
Pump end clearance	0.02-0.07 mm (0.001 -0.003 in)	0.10 mm (0.004 in)



TORQUE VALUES

Engine oil drain plug	Frame side	24-30 Nm (2.4-3.0 kg.m, 17-22 ft.lb)
	Engine side	30-40 N.m (3.0-4.0 kg.m, 22-29 ft.lb)
Oil filter		15-20 Nm (1.5-2.0 kg.m, 11-14 ft.lb)
Main gallery plug		25-35 Nm (2.5-3.5 kg.m, 18-25 ft.lb)
Oil hose (Engine-to-frame)		24-30 Nm (2.4-3.0 kg.m, 17-22 ft.lb)
Oil cooler hose		10-15 Nm (1.0-1.5 kg.m, 7-11 ft.lb)

TOOLS

Special

Oil pressure gauge attachment	07510-MJ10100
Oil filter wrench	07912-6110001

TROUBLESHOOTING

Oil level too low

1. External oil leaks
2. Worn piston rings
3. Worn valve guide or seal

Oil contamination

1. Oil or filter not changed often enough
2. Head gasket faulty
3. Worn piston rings

Low oil pressure

1. Oil level low
2. Pressure relief valve stuck open
3. Plugged oil pick-up screen
4. Oil pump worn
5. External oil leaks

High oil pressure

1. Pressure relief valve stuck closed
2. Plugged oil filter, gallery, or metering orifice
3. Incorrect oil being used

No oil pressure

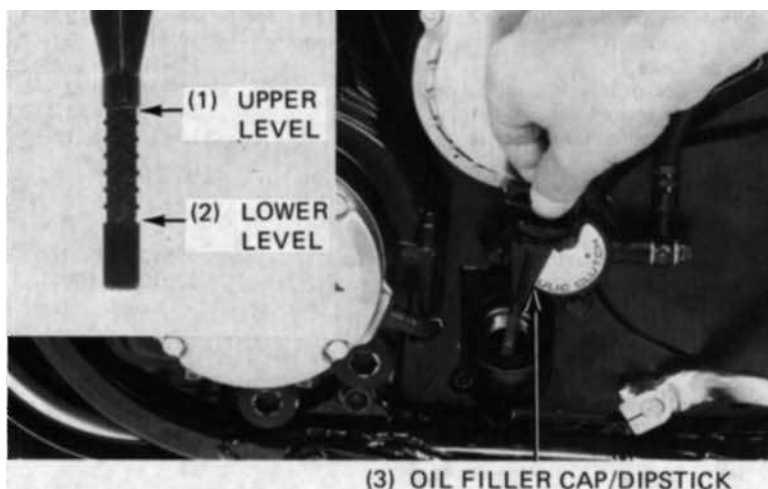
1. Oil level low
2. Oil pump faulty
3. Internal oil leakage



ENGINE OIL LEVEL

Put the motorcycle on its centre stand on level ground. Start the engine and let it idle for 2-3 minutes. Turn off the engine and wait for 2-3 minutes. Remove the filler cap/dipstick, wipe it clean and insert it without screwing it in. Remove the filler cap/dipstick and check the oil level.

If the oil level is below or near the lower level mark on the dipstick, fill to the upper level mark with recommended oil.



ENGINE OIL & FILTER CHANGE

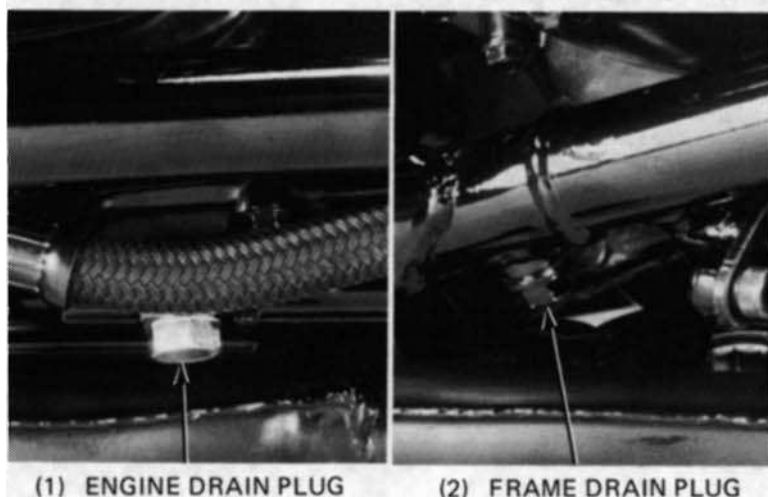
NOTE:

Change engine oil with the engine warm and the motorcycle on its centre stand to assure complete and rapid draining.

WARNING

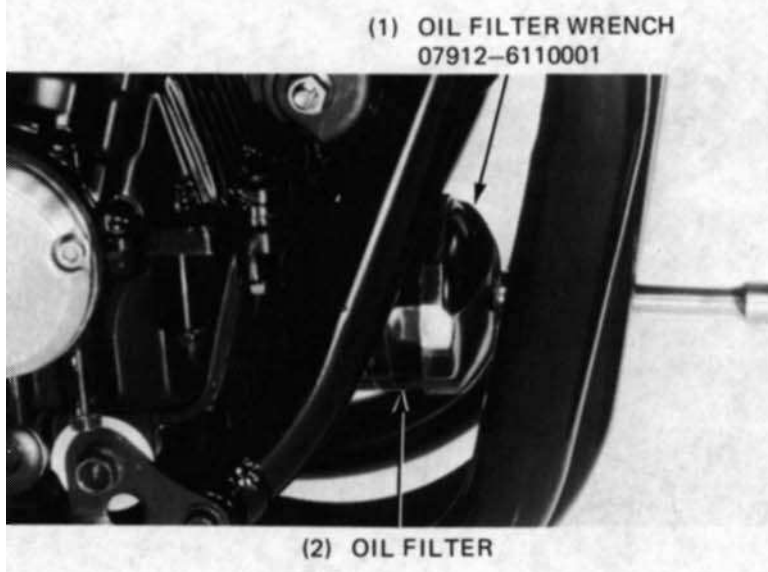
Do not touch the exhaust pipes and frame while they are hot.

Remove the oil filler cap.
Remove the under cowl.
Remove the engine and frame drain plugs, and drain the engine oil.



Loosen the oil filter with a filter wrench (07912-6110001) and remove the filter through the space between the No. 4 exhaust pipe and frame. Discard the oil filter. Apply thread lock agent to the filter boss threads in the crankcase. Coat a new oil filter gasket with clean oil, and install a new oil filter. Tighten the oil filter with the filter wrench (07912-6110001).

TORQUE: 15-20 Nm
(1.5-2.0 kg.m, 11-14 ft.lb)



Wipe spilled oil off the exhaust pipes.



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Check that the sealing washers on the drain plugs are in good condition and install the plugs.

TORQUE:

ENGINE DRAIN PLUG:

30-40 Nm (3.0-4.0 kg.m, 22-29 ft.lb)

FRAME DRAIN PLUG:

24-30 Nm (2.4-3.0 kg.m, 17-22 ft.lb)

Fill the crankcase with 2.8 litres (3.0 U.S. qt, 2.5 Imp. qt) of the recommended oil (page 2-1). Reinstall the oil filler cap/dipstick. Start the engine and let it idle for 2-3 minutes, stop the engine and wait for 2-3 minutes.

Make sure that the oil level is at the upper level mark on the dipstick.

Make sure that there are no oil leaks.

OIL PRESSURE CHECK

Warm the engine up to normal operating temperature (approximately 80° C/176° F). Stop the engine.

Place the motorcycle on its side stand.

Remove the right main gallery plug and connect an oil pressure gauge to the plug hole with an attachment.

Place the motorcycle on its centre stand.

Check the oil level.

Start the engine and check the oil pressure at 6,000 rpm.

OIL PRESSURE: 630kPa(6.3kg/cm², 90psi) at 6,000 rpm (80° C/176° F)

Stop the engine and place the motorcycle on its side stand. Remove the pressure gauge and attachment.

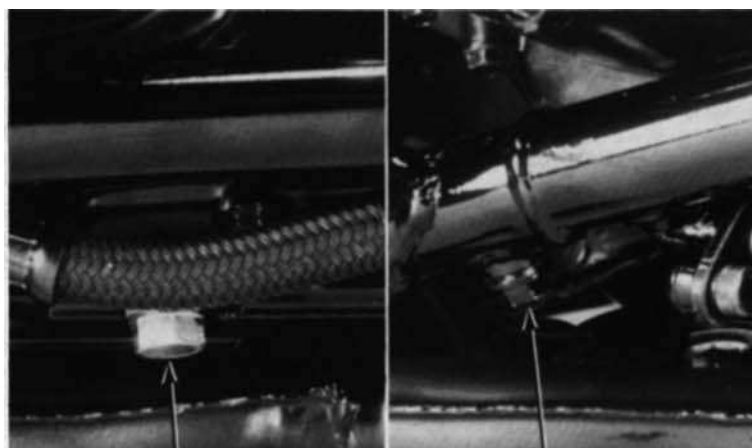
Apply **THREE BOND** sealant or equivalent to the right main gallery plug threads and install the plug with the sealing washer.

TORQUE: 25-35 Nm
(2.5-3.5 kg.m, 18-25 ft.lb)

Start the engine.

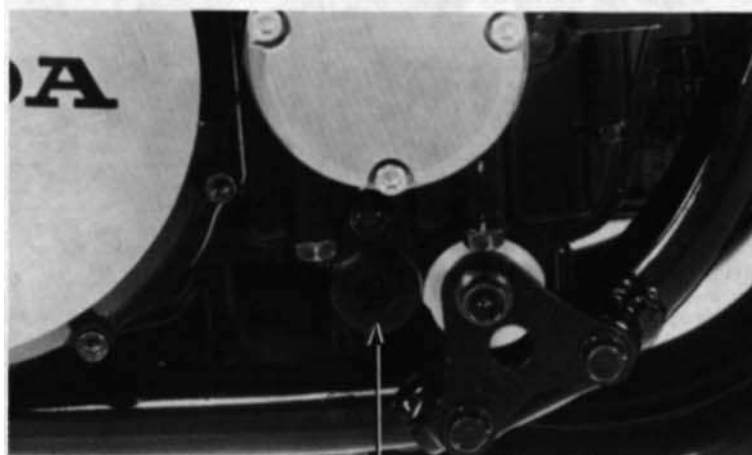
Check that the oil pressure warning indicator goes out after one or two seconds.

If the oil pressure warning indicator stays on, stop the engine immediately and determine the cause.



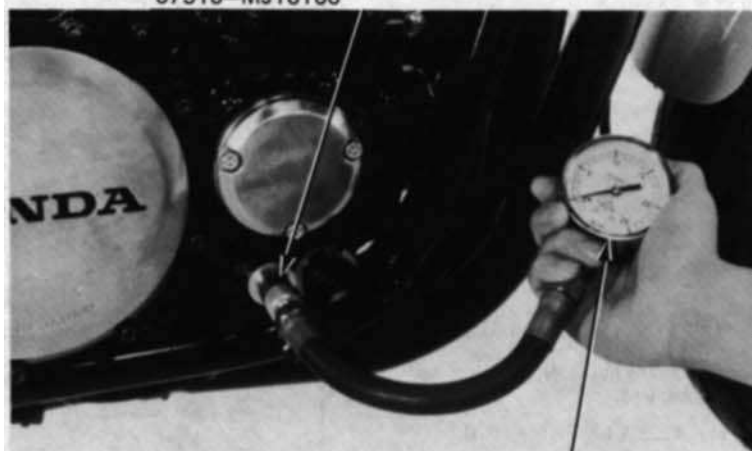
(1) ENGINE DRAIN PLUG

(2) FRAME DRAIN PLUG



(1) RIGHT MAIN GALLERY PLUG

(2) OIL PRESSURE GAUGE ATTACHMENT
07510-MJ10100



(3) OIL PRESSURE GAUGE



OIL STRAINER/PRESSURE REGULATOR VALVE

NOTE:

The oil strainer can be removed with the engine mounted in the frame.

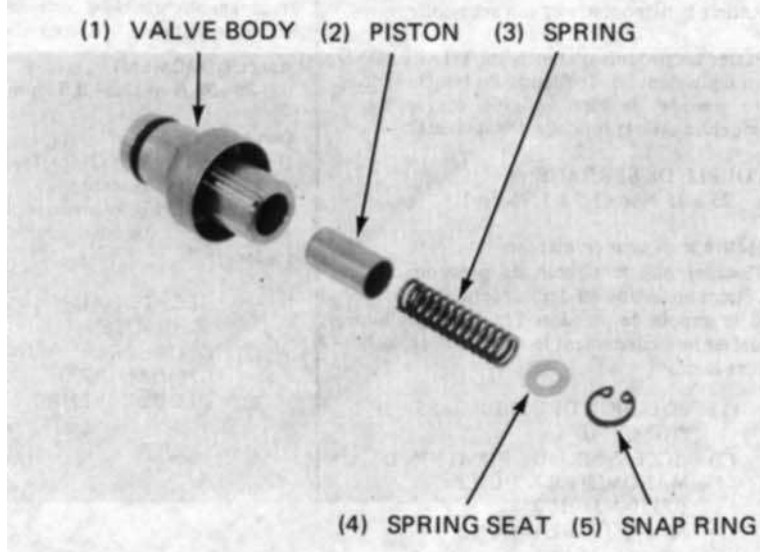
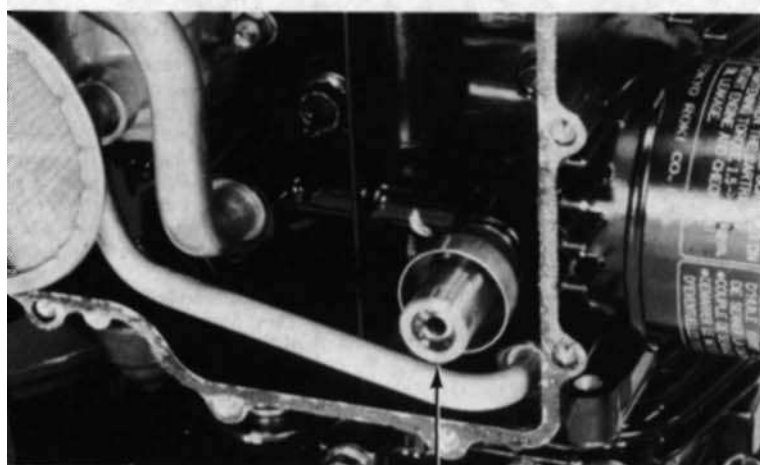
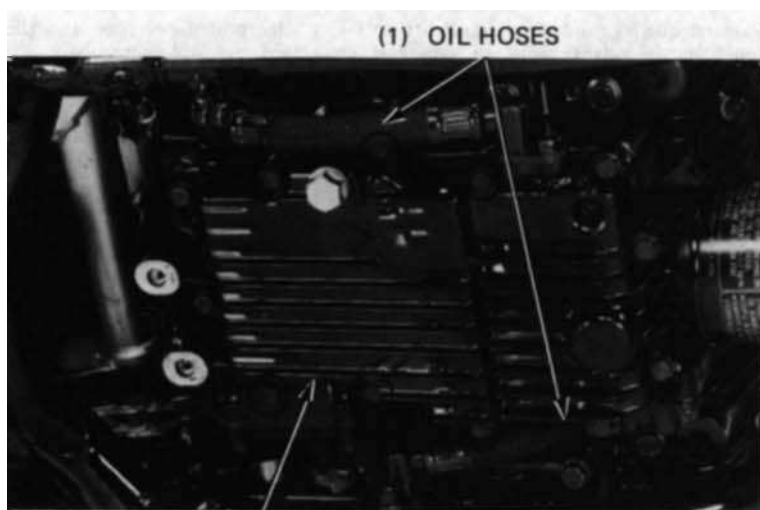
OIL PAN REMOVAL

Remove the under cowl.
Remove the exhaust system.
Remove the oil hoses.
Remove the oil pan and gasket.

RELIEF VALVE INSPECTION

Remove the oil pressure regulator valve and check its operation.

Remove the snap ring, spring seat, spring and piston from the relief valve body. Inspect the piston and inside of the valve body for wear, scratches or scoring. Inspect the spring for wear or damage. Assemble the regulator valve.

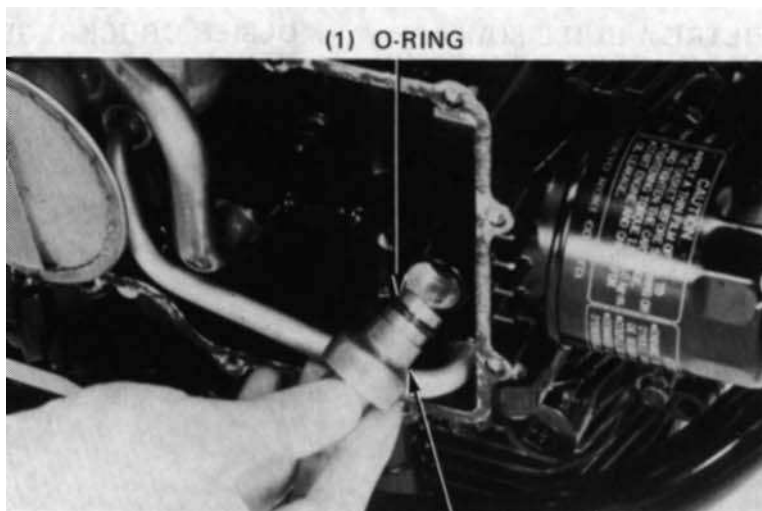




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2. Lubrication

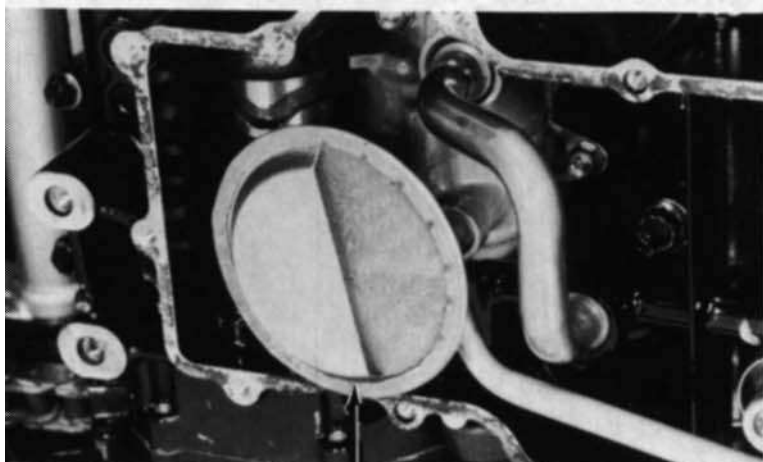
Make sure that the O-ring is good condition. Coat the O-ring with clean oil and install the regulator valve into the crankcase.



(2) REGULATOR VALVE

OIL STRAINER CLEANING

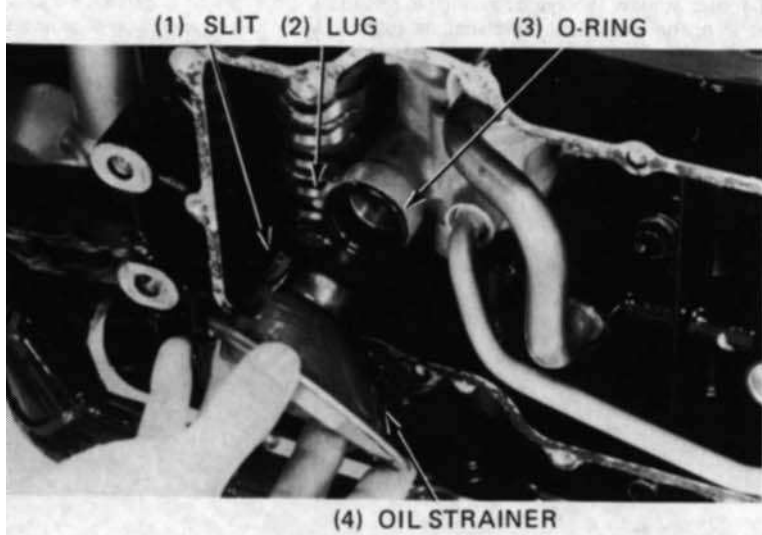
Remove and clean the oil strainer.



(1) OIL STRAINER

Make sure that the strainer O-ring is in good condition.

Coat the O-ring with clean engine oil and install the oil strainer aligning the slit in the strainer with the lug on the oil pump.





OIL PAN INSTALLATION

Install the oil pan with a new gasket.
Tighten the oil pan bolts in criss-cross pattern.
Make sure that the oil hose O-rings are in good condition and install the oil hoses.
Apply thread lock agent to the oil hose bolt threads.
Tighten the oil hose bolts.

TORQUE: **24-30 Nm**
 (2.4-3.0 kg.m, 17-22 ft.lb)

Install the exhaust system and under cowl.
Fill the crankcase with the recommended oil (page 2-1).

OIL PUMP

REMOVAL

NOTE:

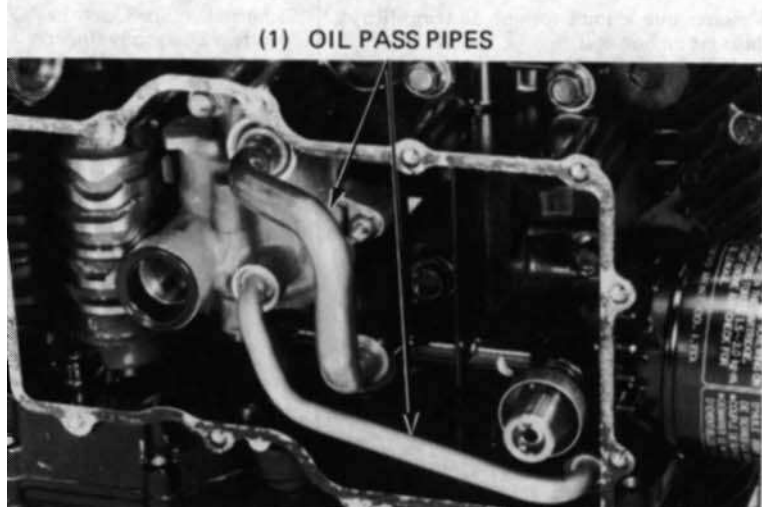
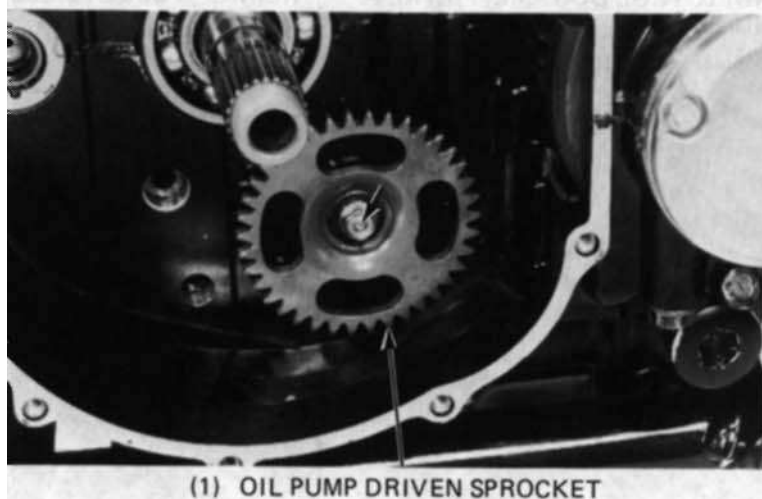
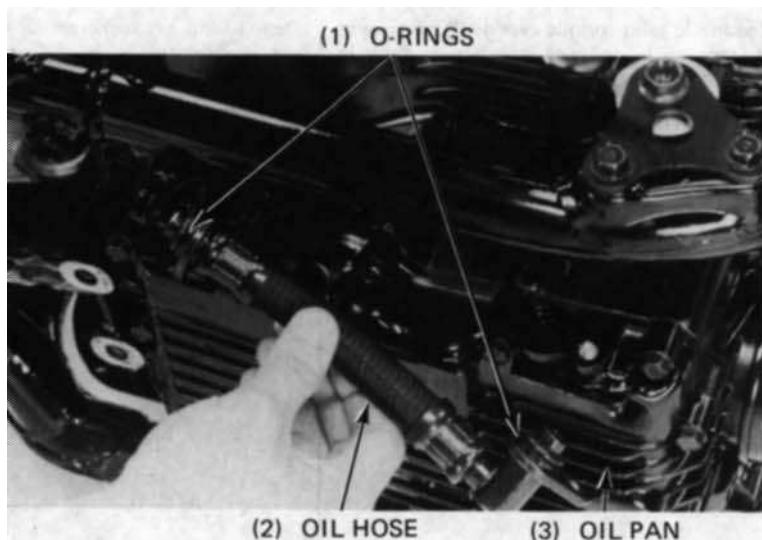
The oil pump can be removed with the engine mounted in the frame.

Remove the following parts:

- under cowl.
- exhaust system.
- clutch assembly.
- oil pan.
- oil strainer.

Remove the oil pump driven sprocket by removing the bolt and washer.

Remove the oil pass pipes.

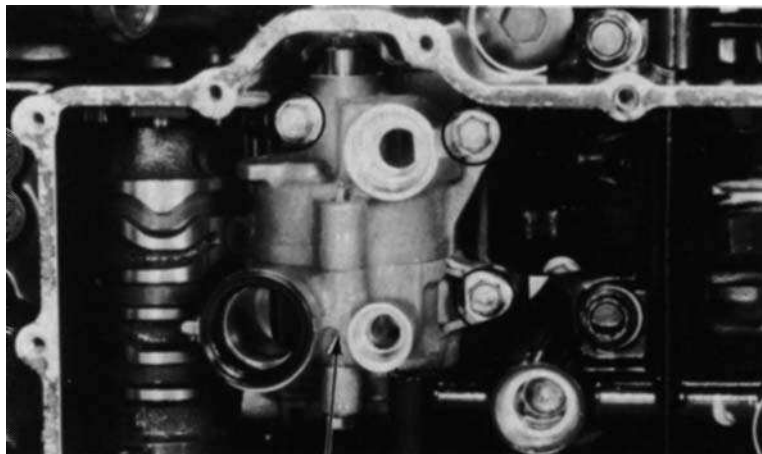




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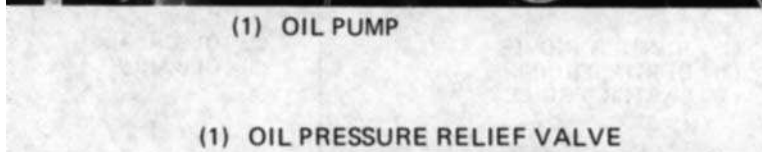
2. Lubrication

Remove the oil pump by removing the three mounting bolts.

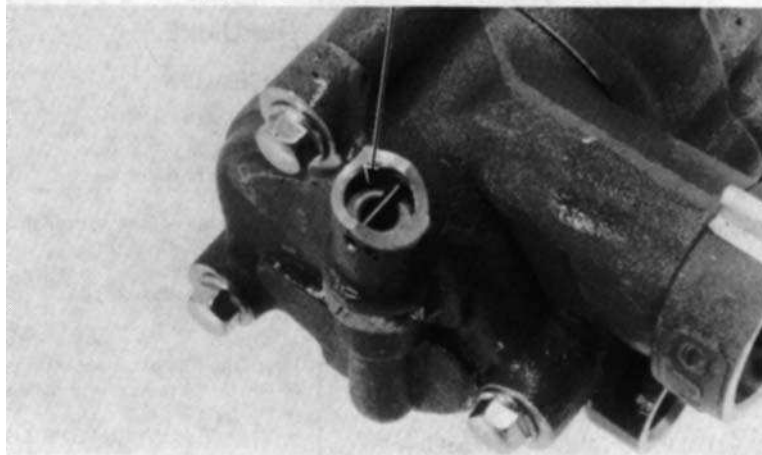


(1) OIL PUMP

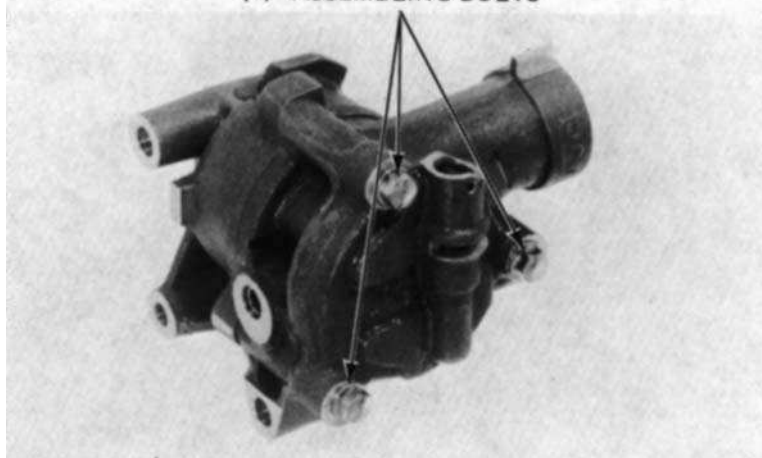
Check the operation of the oil pressure relief valve.



(1) OIL PRESSURE RELIEF VALVE



(1) ASSEMBLING BOLTS



DISASSEMBLY

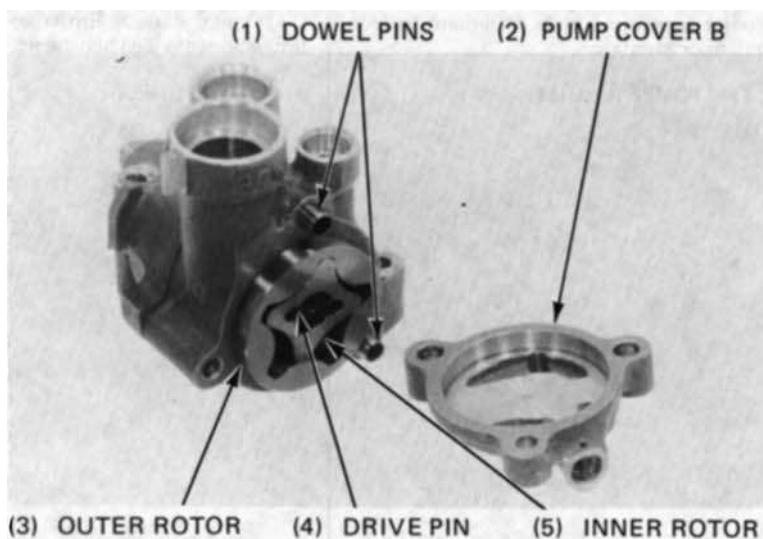
Remove the three bolts assembling the oil pump.



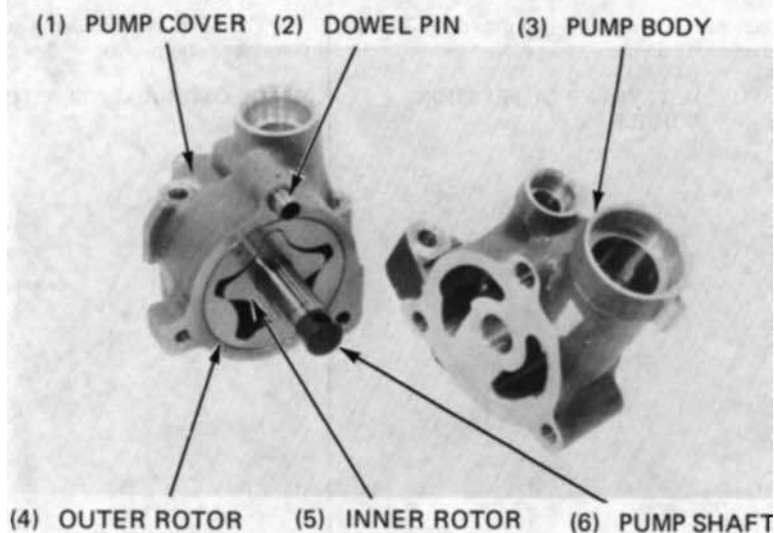
HONDA CBX750F

2. Lubrication

Remove the oil pump cover B.
Remove the dowel pins.
Remove the inner and outer rotors of the cooler pump.
Remove the pump drive pin.



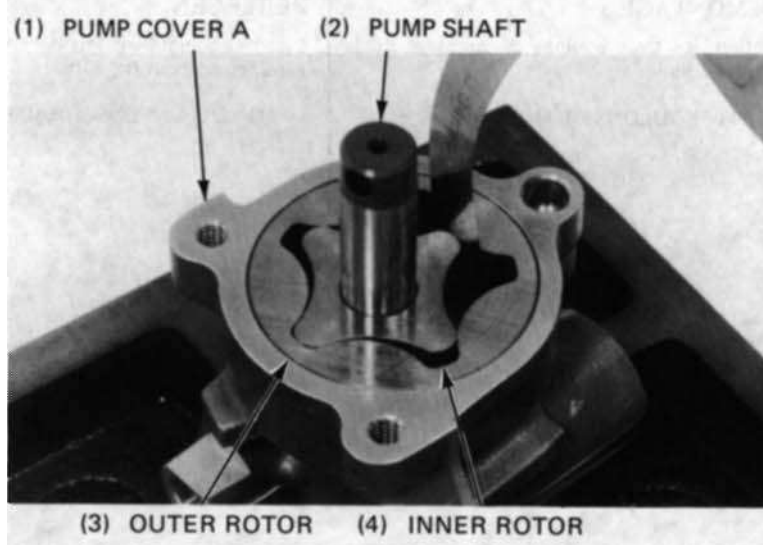
Remove the pump body from pump cover A.
Remove the dowel pin.
Remove the inner and outer rotors of the feed pump.
Remove the pump drive pin and thrust washer.
Remove the oil pump shaft from cover A.



INSPECTION

Install the feed pump inner and outer rotors, thrust washer and pump shaft into pump cover A.
Measure the pump tip clearance.

SERVICE LIMIT: 0.20 mm (0.008 in)



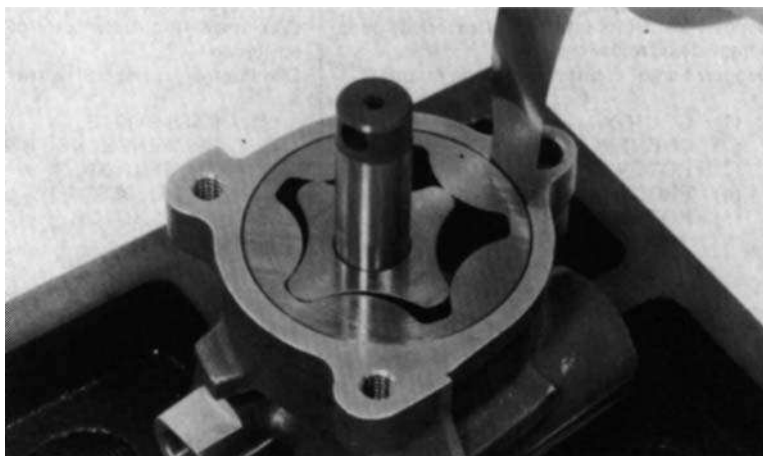


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2. Lubrication

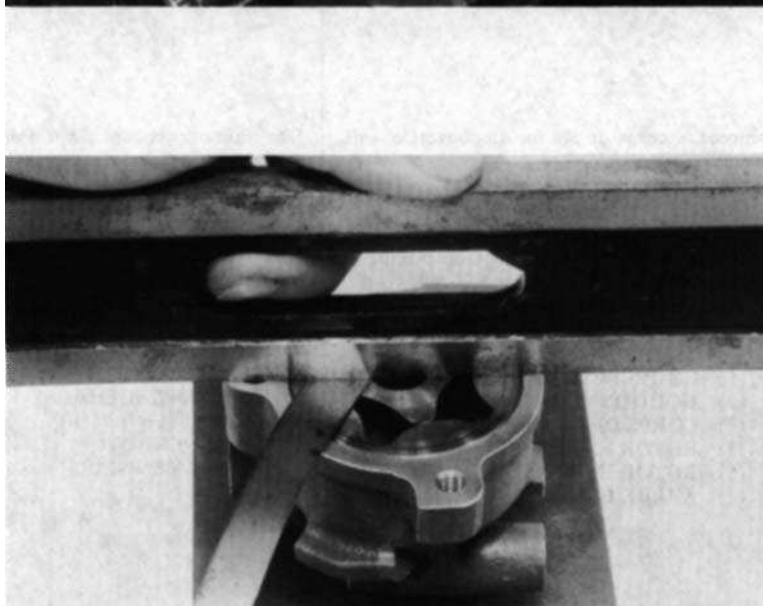
Measure the pump body clearance.

SERVICE LIMIT: 0.35 mm (0.014 in)



Remove the pump shaft and measure the pump end clearance.

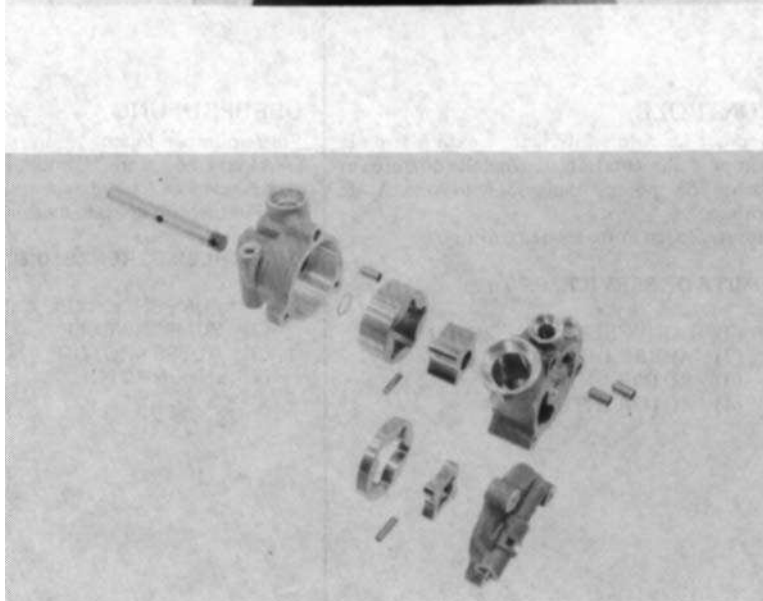
SERVICE LIMIT: 0.10 mm (0.004 in)



ASSEMBLY

Assemble the oil pump in the reverse order of disassembly.

Make sure the oil pump shaft rotates freely after assembling the oil pump.





HONDA CBX750F

2. Lubrication

INSTALLATION

Install the two dowel pins onto the lower crankcase.

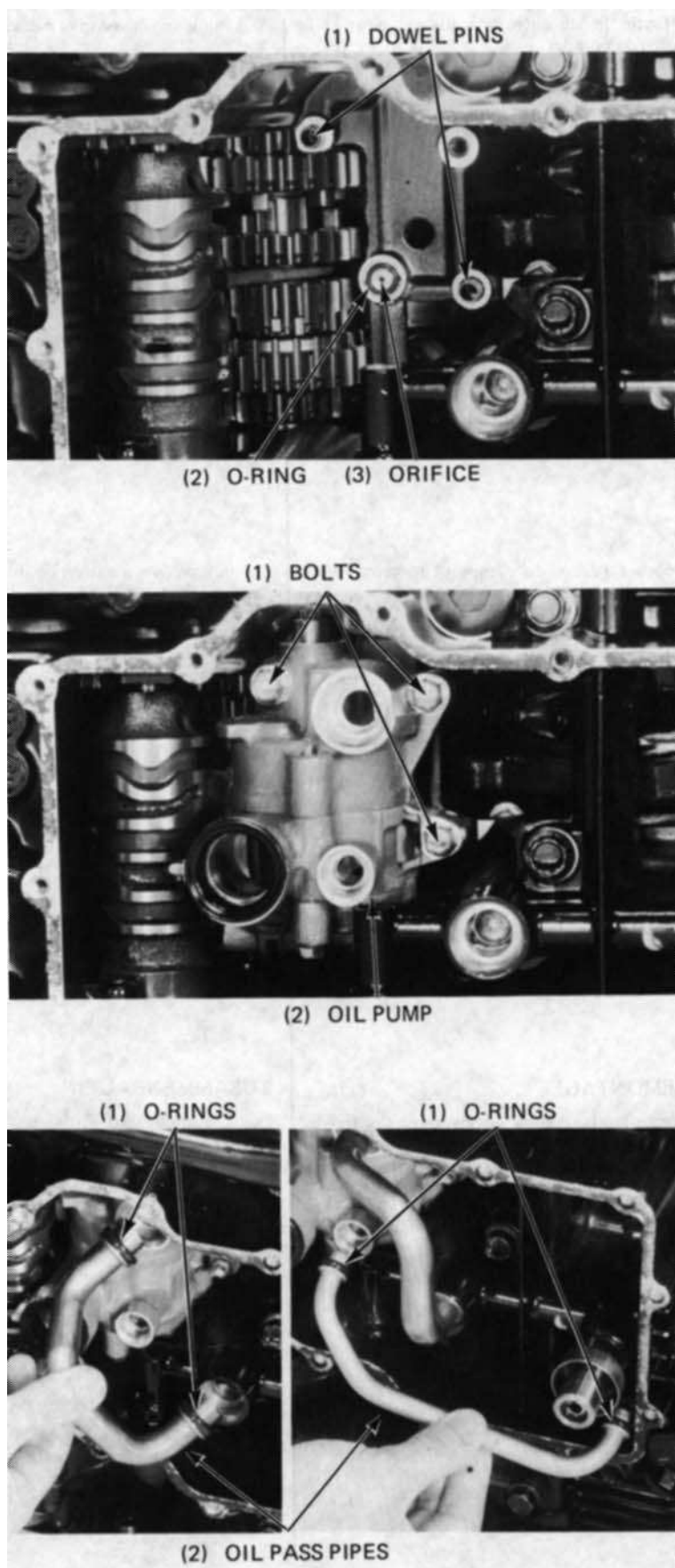
Make sure the orifice is not clogging and the O-ring is in good condition.

Install the orifice and O-ring.

Install the oil pump onto the lower crankcase and tighten the three mounting bolts.

Make sure that the O-rings on the oil pass pipes are in good condition.

Coat O-rings with clean engine oil and install the oil pass pipes.





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2. Lubrication

Install the oil pump driven sprocket with the washer and bolt.

Install the following parts:

- oil strainer.
- oil pan.
- clutch assembly.
- exhaust system.
- under cowl.

Fill the engine with the recommended oil (page 2-1).

OIL COOLER

REMOVAL

Remove the fairing and headlights with the bracket (section 13).

Drain the engine oil (page 2-3).

Disconnect the oil hoses from the cooler by removing the four bolts.

Remove the two mount nuts and the oil cooler.

INSPECTION

Inspect the oil cooler soldered joints and seams for leaks.

Check the air passages for clogging or damage.

Straighten bent fins and collapsed core tubes.

Remove insects, mud or any obstruction with compressed air or low water pressure.

INSTALLATION

Install the oil cooler onto the lower mounts and mount studs properly and secure with the mount nuts.

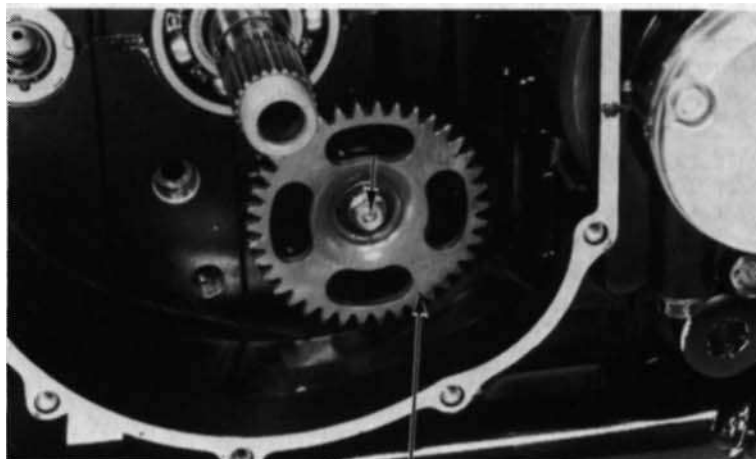
Connect the oil hoses to the cooler and tighten the bolts.

TORQUE: **10-15Nm**
 (1.0-1.5 kg.m, 7-11 ft.lb)

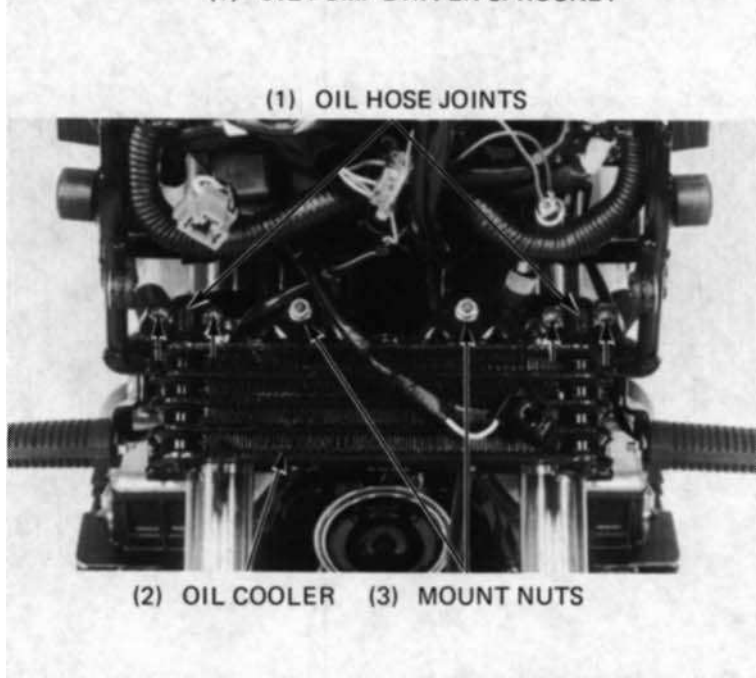
Install the headlight with the bracket, and the fairing.

Fill the crankcase with the recommended oil (page 2-1),

Start the engine and check for oil leaks.

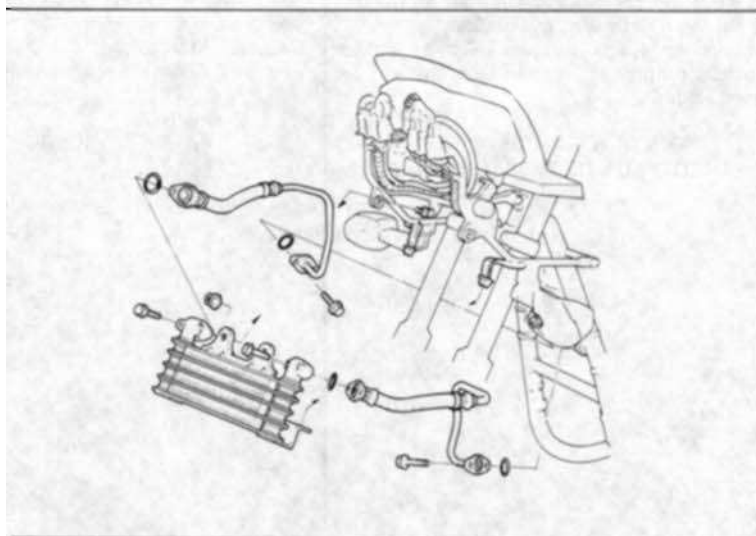


(1) OIL PUMP DRIVEN SPROCKET



(1) OIL HOSE JOINTS

(2) OIL COOLER (3) MOUNT NUTS





CONTROL CABLE LUBRICATION

Periodically, disconnect the throttle cables at their upper ends. Thoroughly lubricate the cables and their pivot points with a commercially available cable lubricant or a light weight oil.

